Recommended Assessment

Forward Kinematics

Workspace Identification

1. How does the range of motion of each joint contribute to the overall workspace shape?

2. Looking at the generated graph, does the workspace match your expectations based on how you moved the arm? Add your generated graph.

3. Can you identify areas where the manipulator could not reach? Why do you think these areas are inaccessible? Add pictures of your generated plots.

Lead Through

4. Attach screenshots from the “joint position tracking” scope. Are the recorded trajectory and measured position Identical? If not, why?

5. Attach screenshots from the “EE position tracking scope”. When comparing the screenshots from the above question, what differences do you notice? Why?

6. Examine the data from the "EE Position Tracking Scope" when guiding the arm along a closed-loop trajectory both near the base and further away. What differences do you notice between the two scenarios? Support your observations with screenshots from the "EE Position Tracking Scope.

7. By manually guiding the manipulator, how easy was it to trace a desired path? How accurately did the end-effector follow the intended trajectory?

8. If you were to program the manipulator to follow the same path instead of guiding it by hand, what challenges would you encounter in ensuring accuracy and repeatability?